

AI/ML Intern Assessment

Objective

Build an **AI-powered Knowledge Base Chat** application that allows users to upload documents and ask questions based on their content.

Requirements

- Allow users to upload one or more PDF documents.
- Generate answers only from the uploaded documents using a Retrieval-Augmented Generation (RAG) approach.
- Display the source document and relevant page/reference for every answer.
- Show a confidence score (or confidence level) for each response.
- If the requested information is not available in the uploaded documents, respond with **"I don't know based on the provided documents."** Do not generate unsupported or hallucinated answers.

Technical Expectations

- Use any LLM of your choice.
- Implement document chunking and embeddings.
- Store embeddings in a vector database.
- Build a clean, intuitive, and user-friendly interface.
- Handle errors gracefully and provide meaningful user feedback.

Deliverables

- Source code hosted on GitHub.
- README with setup and usage instructions.
- `.env.example` file.
- A brief explanation of the architecture and design decisions.
- A short demo video (2–3 minutes).

Evaluation Criteria

- Correct implementation of the RAG pipeline.
- Accuracy and relevance of responses.
- Source citation and retrieval quality.
- Hallucination prevention.
- Code quality and project structure.
- User experience and documentation.